



**RAJALAKSHMI
ENGINEERING COLLEGE**

An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

***DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING LAB MANUAL***

CS23432 – Software Construction

(REGULATION 2023)

***RAJALAKSHMI ENGINEERING COLLEGE
Thandalam, Chennai-602015***

Name:

Register No:

Year / Branch / Section:

Semester:

Academic Year:

INDEX

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1.		Azure Devops Environment Setup.
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7.		Designing Architectural and ER Diagrams for Project Structure.
8.		Testing – Test Plans and Test Cases.
9.		Load Testing and Pipelines.
10.		GitHub: Project Structure & Naming Conventions.

EXP NO: 1

AZURE DEVOPS ENVIRONMENT SETUP

Aim:

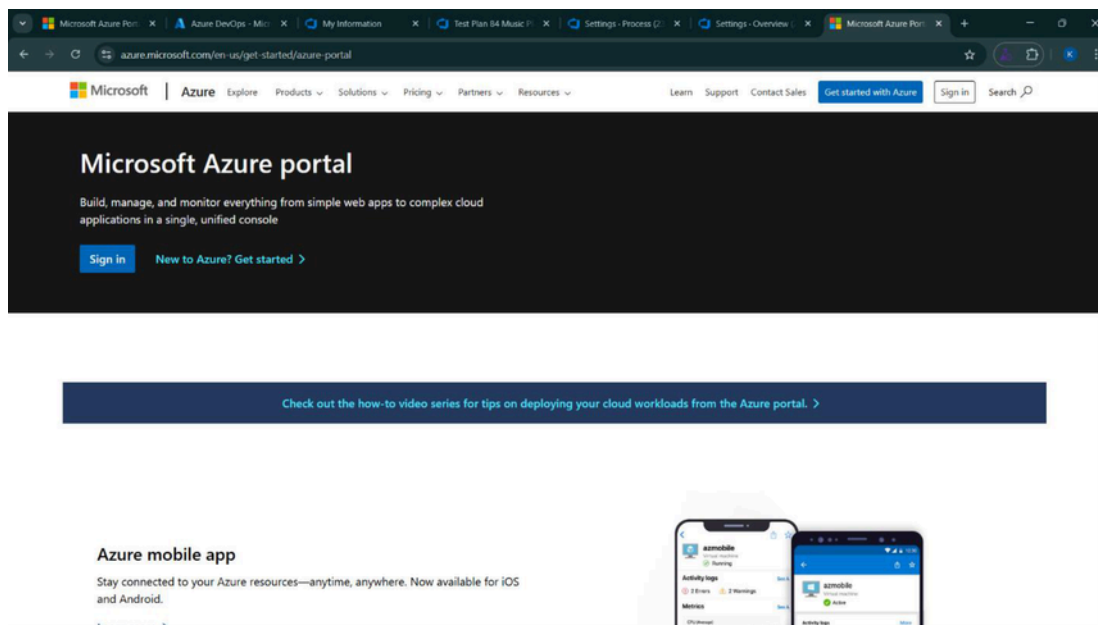
To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

INSTALLATION

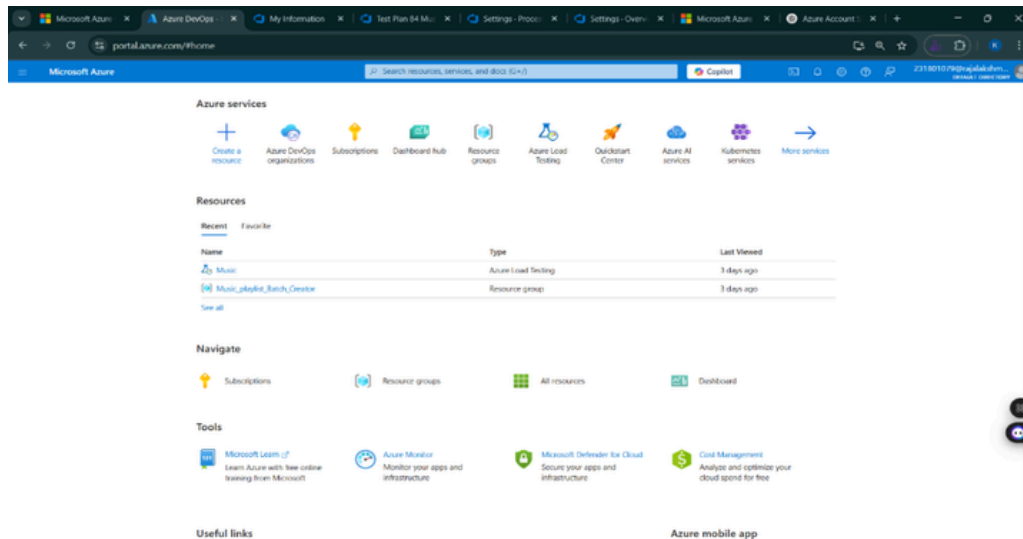
1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

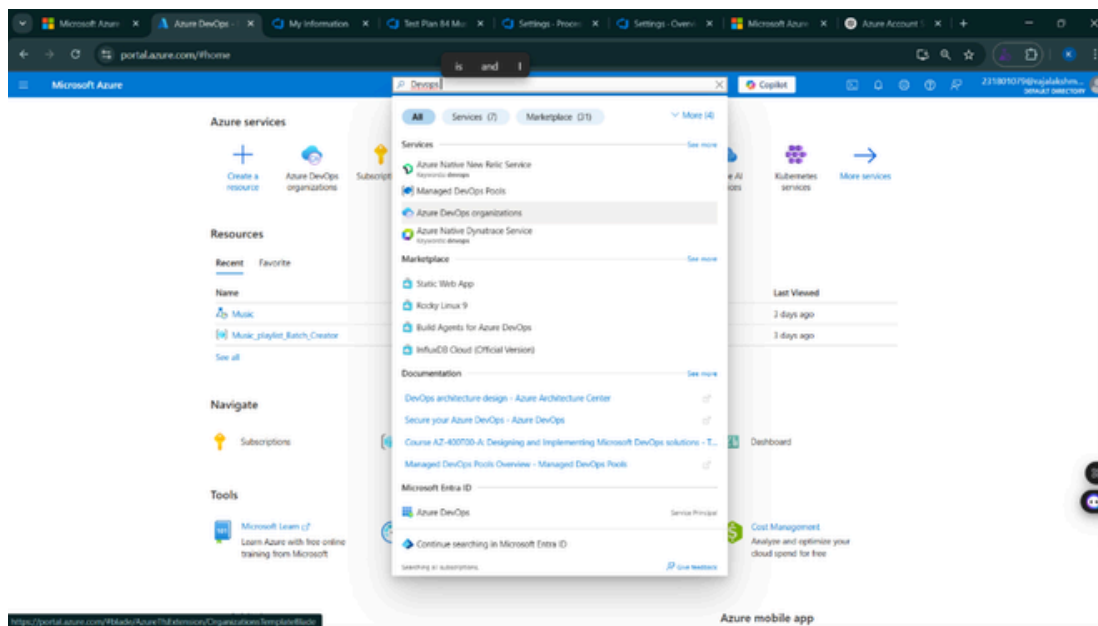
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



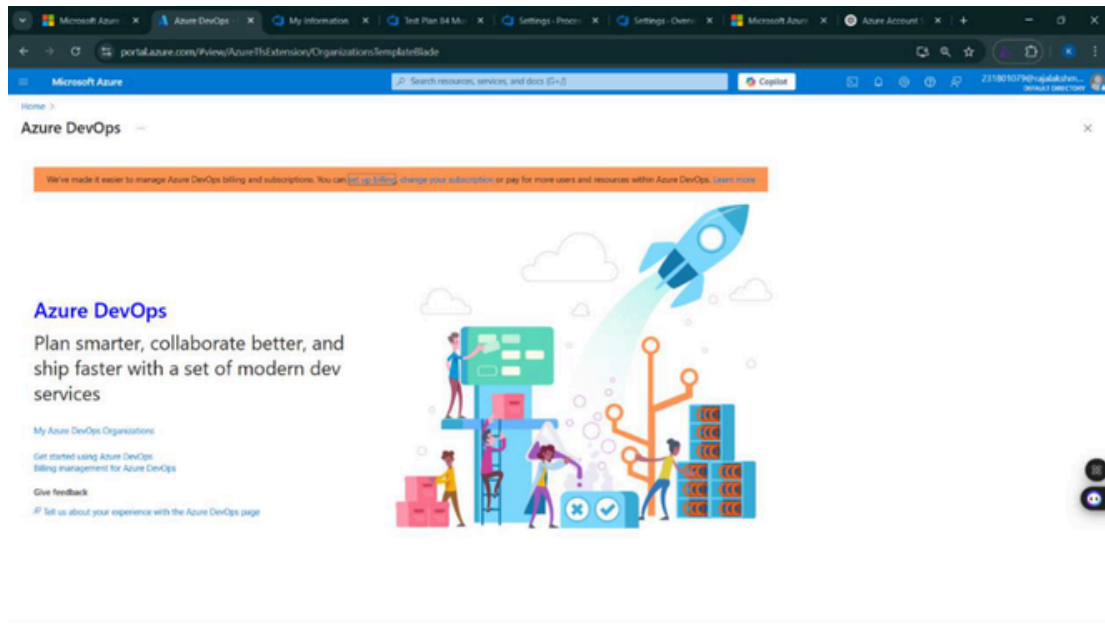
2. Azure home page



3. Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.



4. Click on the *My Azure DevOps Organization* link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azureportal.

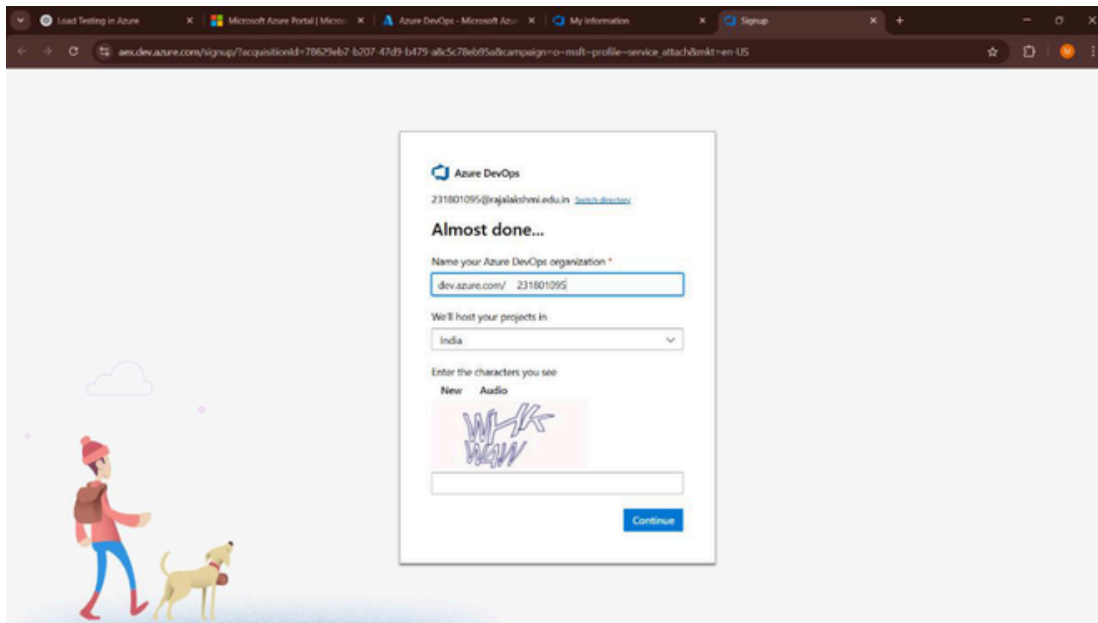
EXP NO: 2

AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

1. Create An Azure Account



2. Create the First Project in Your Organization

- a. After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- b. On the organization's **Home page**, click on the **New Project** button.
- c. Enter the project name, description, and visibility options:
 - Name:** Choose a name for the project (e.g., **LMS**).
 - Description:** Optionally, add a description to provide more context about the project.
 - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- d. Once you've filled out the details, click **Create** to set up your first project.

Create new project



Project name *

Description

Visibility

 **Private**

Only people you give access to will be able to view this. Want to create a public project? [Try GitHub](#)

▼ Advanced

Cancel

Create

3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

Microsoft

Suriya prakashSign out

SP

Suriya prakash

suriyaprakasha.2024.aiml@rajalakshmi.edu.in

Microsoft account

India

suriyaprakasha.2024.aiml@rajalakshmi.edu.in

Visual Studio Dev Essentials

Get everything you need to build and deploy your app on any platform.

Use your benefits

Azure DevOps Organizations

Create new organization

dev.azure.com/suriyaprakasha2024aiml (Owner)

Projects

school organization

online pharmacy management system

New project

Actions

Open in Visual Studio

dev.azure.com/harrishs2024aiml (Member)

4.Project dashboard

DevOps

suriyaprakasha2024aiml / school organization / Overview / Summary

Search

Private

Invite

ol organization

view

mary

boards

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s

ines

Plans

icts

ct settings

SO school organization

About this project

Like 0

school organization project is a project to manage the school activities monitoring school attendance grades and managing the tasks that authority needs

Project stats

Period: Last 7 days

Boards

0 Work Items

0 Work Items

Members 1

SP

STUDENT REG NO

CS23432

5.To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

DevOps suriyaprakash2024aiml / school organization / Boards / Work items

Search

Work items

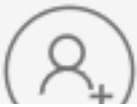
Recently updated + New Work Item Open in Queries Column Options Import Work Items Recycle Bin

Filter by keyword

ID	Title	Assigned To	State	Area Path	Tags
11	marks management	Unassigned	New	school organization	
10	marks management	Suriya prakash	New	school organization	
6	attendance management	Suriya prakash	New	school organization	
5	attendance monitoring	Unassigned	New	school organization	
2	organizer	Suriya prakash	To Do	school organization	
1	organizer	Unassigned	To Do	school organization	

Microsoft Sign out

 **Suriya prakash**
suriyaprakash.a.2024.aiml@raj...
[My Microsoft account](#)
[Switch directory](#) ...

 Sign in with a different account

Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

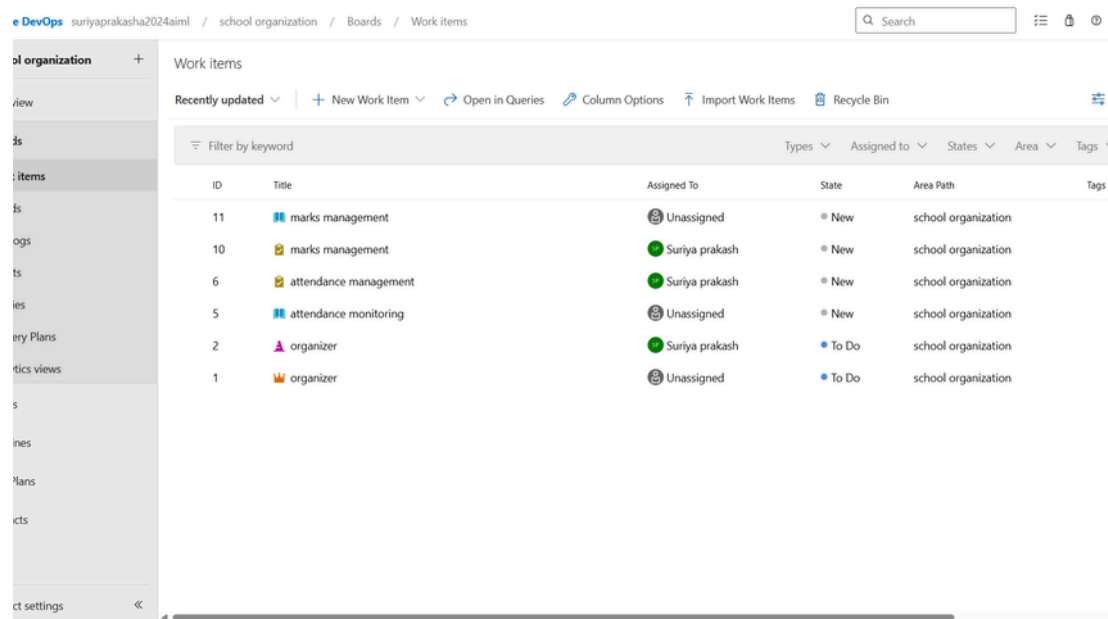
EXP NO: 3

SETTINGUPEPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To learn about how to create epics, user story, features, backlogs for your assigned project.

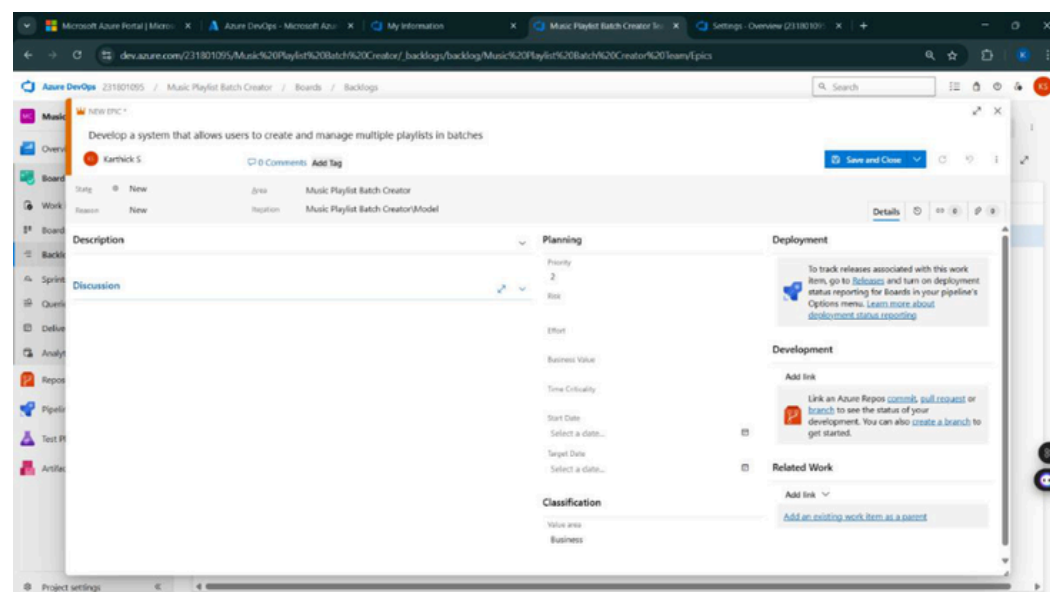
Create Epic, Features, User Stories, Task



The screenshot shows the 'Work Items' page in Azure DevOps for the project 'school organization'. The page has a search bar and navigation links like 'New Work Item', 'Open in Queries', 'Column Options', 'Import Work Items', and 'Recycle Bin'. A table lists work items with columns for ID, Title, Assigned To, State, Area Path, and Tags.

ID	Title	Assigned To	State	Area Path	Tags
11	marks management	Unassigned	New	school organization	
10	marks management	Suriya prakash	New	school organization	
6	attendance management	Suriya prakash	New	school organization	
5	attendance monitoring	Unassigned	New	school organization	
2	organizer	Suriya prakash	To Do	school organization	
1	organizer	Unassigned	To Do	school organization	

1.Fill in Epics



2.Fill in Features

The screenshot shows the Azure DevOps web interface. The browser address bar displays the URL: `dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_backlog/backlog/Music%20Playlist%20Batch%20Creator%20Team/Epics?workItem=26`. The page title is "Music Playlist Batch Creator / Boards / Backlogs". The left sidebar shows the "Music" team and various work item types. The main content area displays a Feature work item (ID 26) titled "Auto-Playlist Creation Based on user preference". The work item is created by Karthick S. and is currently in the "New" state. The description field is empty, with a placeholder "Click to add Description". The "Discussion" section is also empty, with a placeholder "Add a comment. Use # to link a work item, @ to mention a person, or / to link a pull request." The "Planning" section shows fields for Priority (2), Risk, Effort, Business Value, Time Criticality, Start Date, and Target Date. The "Classification" section shows Value area (Business). The "Deployment" section contains instructions on how to track releases associated with this work item. The "Development" section contains instructions on how to link an Azure Repos commit, pull request, or branch to see the status of your development. The "Related Work" section shows a parent work item (ID 47) titled "As a user, I should be able to create playlists based on various categories (like mood, genre, artist, popularity, lyrics) as I need".

3.Fill in User Story Details

The screenshot shows the Azure DevOps web interface. The browser address bar displays the URL: `dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_backlog/backlog/Music%20Playlist%20Batch%20Creator%20Team/Epics?workItem=47`. The page title is "Music Playlist Batch Creator / Boards / Backlogs". The left sidebar shows the "Music" team and various work item types. The main content area displays a User Story work item (ID 47) titled "As a user, I should be able to create playlists based on various categories (like mood, genre, artist, popularity, lyrics) as I need". The work item is created by Karthick S. and is currently in the "Resolved" state. The description field is empty, with a placeholder "Click to add Description". The "Acceptance Criteria" section is also empty, with a placeholder "Click to add Acceptance Criteria". The "Discussion" section is empty, with a placeholder "Add a comment. Use # to link a work item, @ to mention a person, or / to link a pull request." The "Planning" section shows fields for Story Points (3), Priority (2), and Risk. The "Classification" section shows Value area (Business). The "Deployment" section contains instructions on how to track releases associated with this work item. The "Development" section contains instructions on how to link an Azure Repos commit, pull request, or branch to see the status of your development. The "Related Work" section shows a parent work item (ID 26) titled "Auto-Playlist Creation Based on user preference".

Result:

Thus, the creation of epics, features, user story and task has been created successfully.

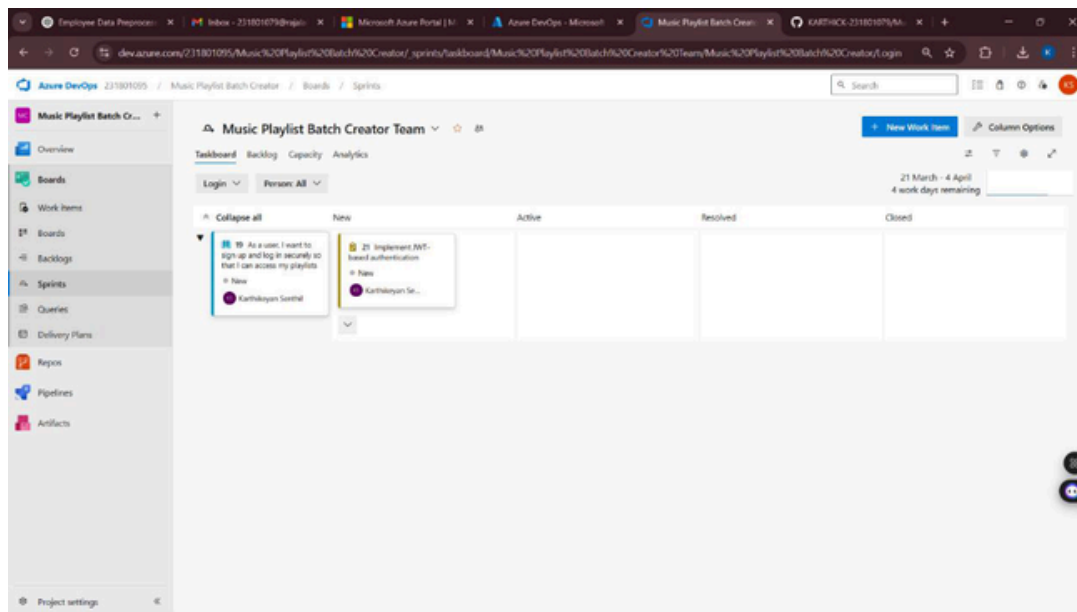
EXP NO: 4

SPRINT PLANNING

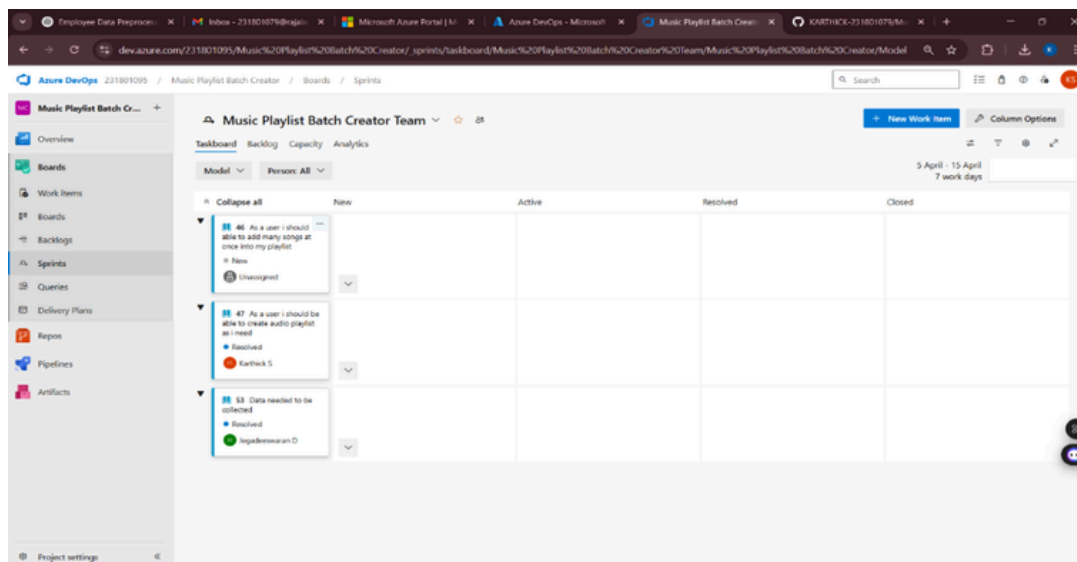
Aim:

To assign user story to specific sprint for the Music Playlist Batch Creator Project.

Sprint Planning Sprint 1



Sprint 2



Sprint 3

The screenshot shows the Azure DevOps Sprints board for the 'Music Playlist Batch Creator Team'. The board is titled 'Sprint 3' and shows a timeline from 15 April to 30 April, lasting 12 work days. The board is divided into columns: 'None', 'Active', 'Resolved', and 'Closed'. There are four work items in the 'None' column, all assigned to 'Karthick S.'. The work items are:

- 48: As a user, I can Connect with my music app
- 38: As a user, I want to link my Spotify account or any music platform so that I can import my music preferences
- 49: As a user I need to have real time meta data
- 58: As a user I need see my playlist in one place

Sprint 4

The screenshot shows the Azure DevOps Sprints board for the 'Music Playlist Batch Creator Team'. The board is titled 'Sprint 4' and shows a timeline from 29 April to 30 April, lasting 2 work days. The board is divided into columns: 'None', 'Active', 'Resolved', and 'Closed'. There are three work items in the 'None' column, all assigned to 'Karthick S.'. The work items are:

- 23: As a user I can able to customize my playlist
- 30: As a user I should be able to rename record and change the playlist
- 31: As a user I should be able to access a friendly and Modern UI

Result:

The Sprints are created for the Music Playlist Batch Creator Project.

EXP NO: 5

POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories - Music Playlist Batch Creator Project.

PokerEstimation

Employee Data Preprocess | Inbox - 231801079@raj... | Microsoft Azure Portal | Azure DevOps - Microsoft | Music Playlist Batch Cre... | KARTHICK-231801079/M... | +

dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_backlog/backlog/Music%20Playlist%20Batch%20Creator%20Teams/Epics?workitem=47

USER STORY 47:
47 As a user i should be able to create audio playlist as i need

Karthick S 0 Comments Add Tag Save and Close Follow

Status: Resolved Area: Music Playlist Batch Creator Updated by Malhu karthick Satagi Reddy: Feb 18
Reason: Code complete and unit t Iteration: Music Playlist Batch Creator\Model Details

Description
Click to add Description.

Acceptance Criteria
Click to add Acceptance Criteria.

Discussion
Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.
switch to Markdown editor

Planning
Story Points: 3
Priority: 2
Risk:

Classification
Value area: Business

Deployment
To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting.

Development
Add link
Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.

Related Work
Add link
Parent
26 Auto-Playlist Creation Based on user preference
Updated Feb 18 @ New

Result:

The Estimation/Story Points is created for the project using Poker Estimation.

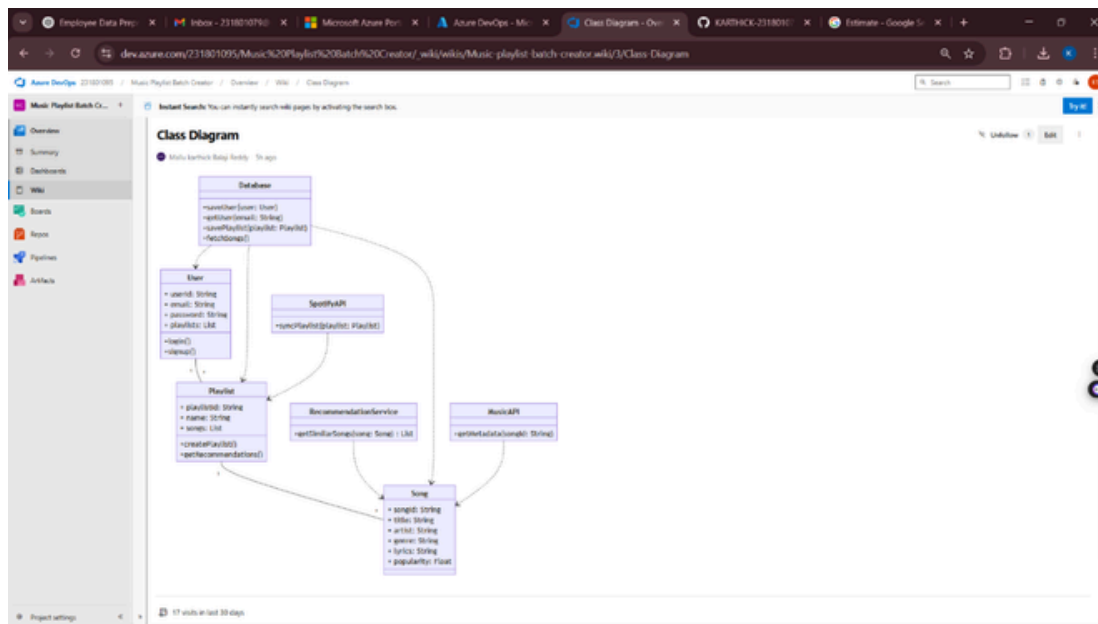
EXP NO: 6

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

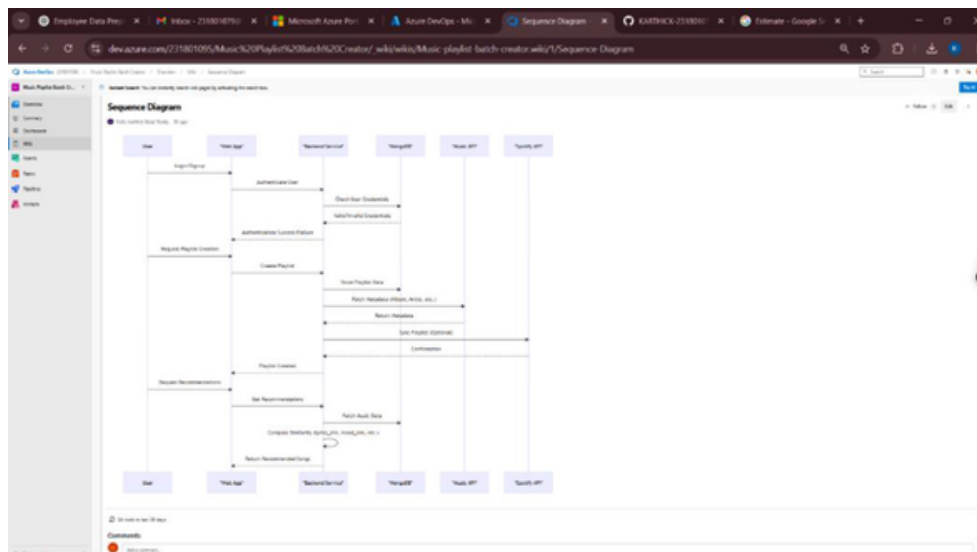
Aim:

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. Class Diagram



6B. Sequence Diagram



Result:

The Class Diagram and Sequence Diagram is designed Successfully for the Music Playlist Batch Creator.

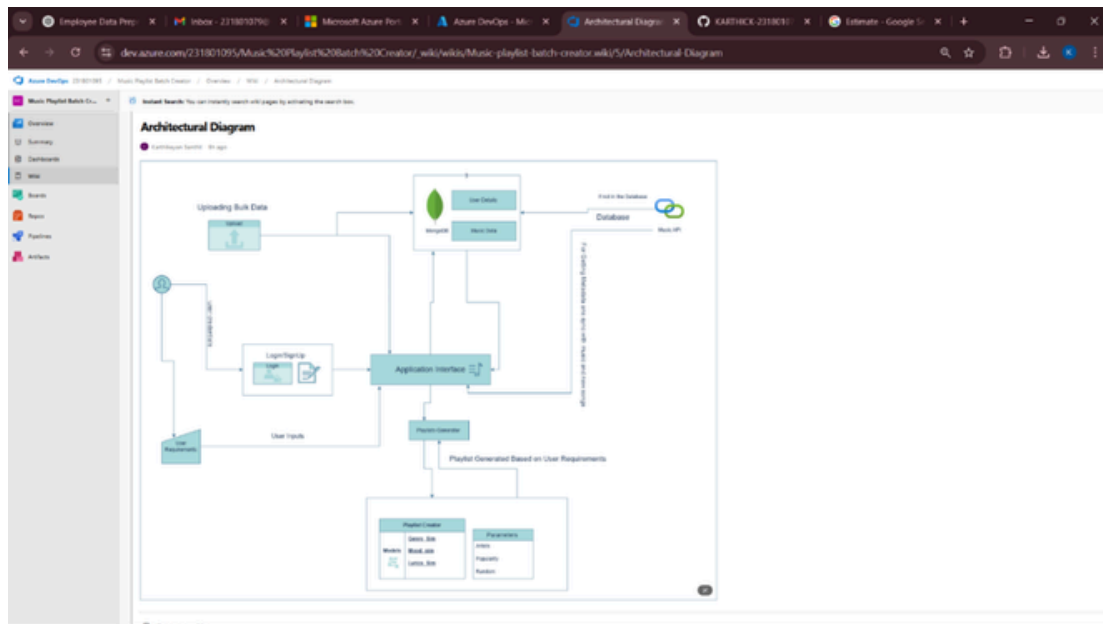
EXP NO: 7

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

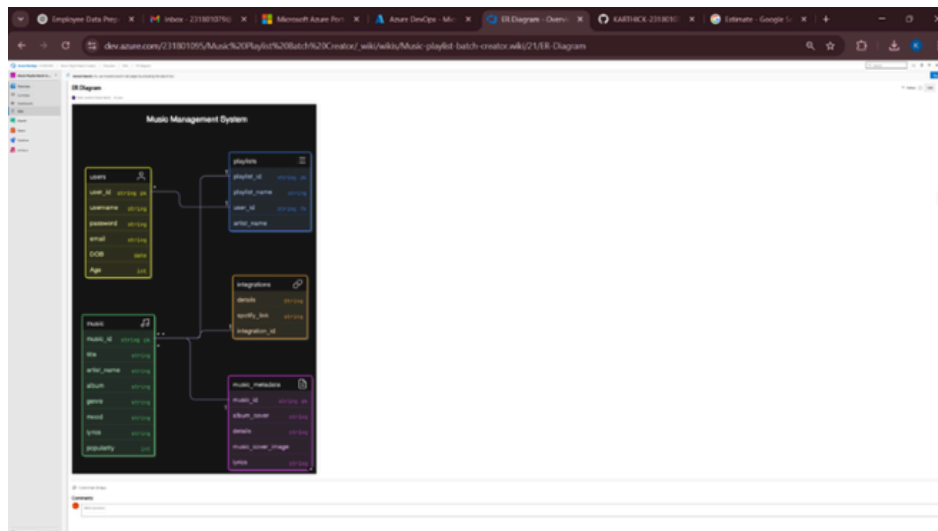
Aim:

To Design an Architectural Diagram and ER Diagram for the given Project.

7A. Architectural Diagram



7B.ER Diagram



Result:

The Architecture Diagram and ER Diagram is designed Successfully for the Music Playlist Batch Creator

EXP NO: 8

TESTING – TEST PLANS AND TEST CASES

Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case

Test Case Design Procedure

1. Understand Core Features of the Application

- o User Signup & Login
- o Viewing and Managing Playlists
- o Fetching Real-time Metadata
- o Editing playlists (rename, reorder, record)
- o Creating smart audio playlists based on categories (mood, genre, artist, etc.)

2. Define User Interactions

- o Each test case simulates a real user behaviour (e.g., logging in, renaming a playlist, adding a song).

3. Design Happy Path Test Cases

- o Focused on validating that all features function as expected under normal conditions.
- o Example: User logs in successfully, adds item to playlist, or creates a category-based playlist.

4. Design Error Path Test Cases

- o Simulate negative or unexpected scenarios to test robustness and error handling.
- o Example: Login fails with invalid credentials, save fails when offline, no recommendations found.

5. Break Down Steps and Expected Results

- o Each test case contains step-by-step actions and a corresponding expected outcome.
- o Ensures clarity for both testers and automation scripts.

6. Use Clear Naming and IDs

- o Test cases are named clearly (e.g., TC01 – Successful Login, TC10 – Save Playlist Fails).
- o Helps in quick identification and linking to user stories or features.

7. Separate Test Suites

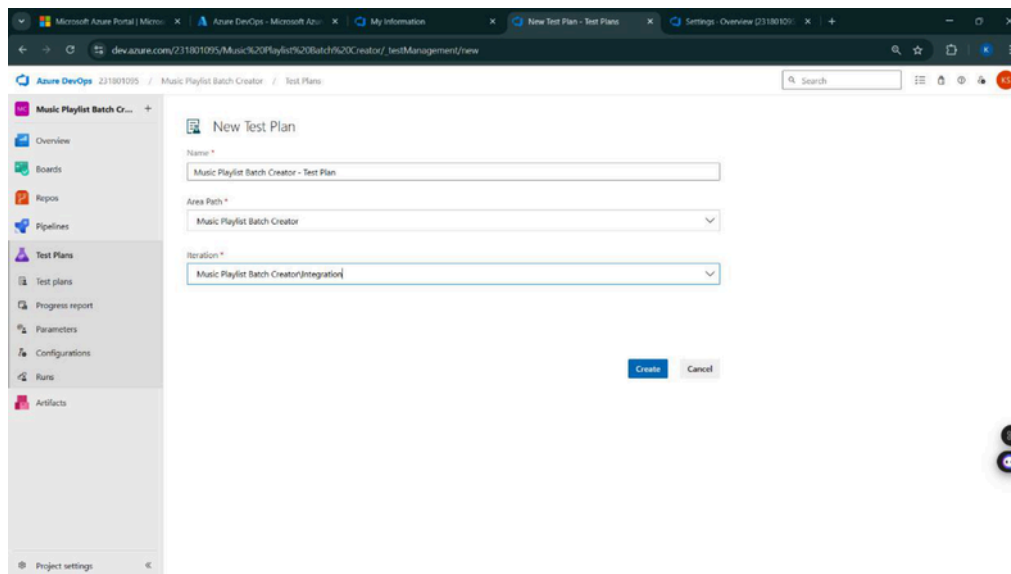
- o Grouped test cases based on functionality (e.g., Login, Playlist Editing, Recommendation System).

- o Improves organization and test execution flow in Azure DevOps.

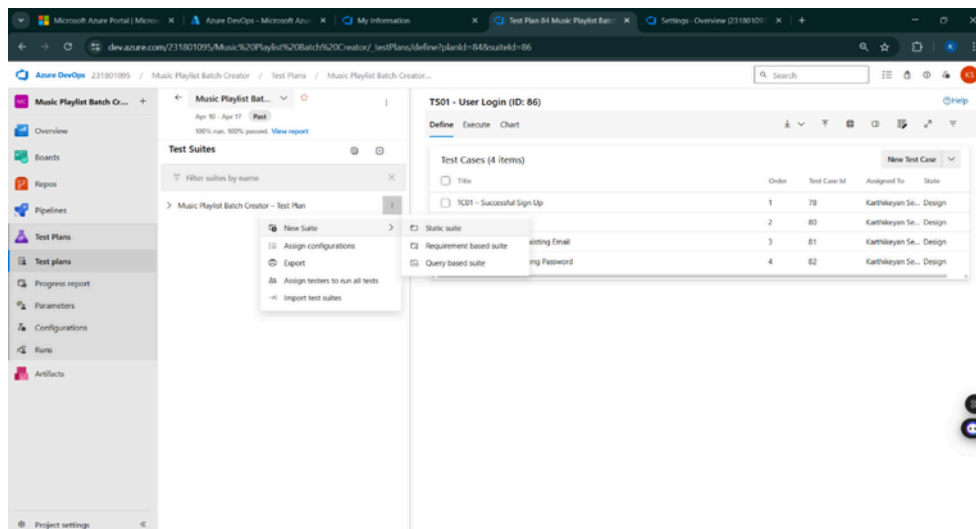
8. *Prioritize and Review*

- o Critical user actions are marked high-priority.
- o Reviewed for completeness and traceability against feature requirements.

1. *New test plan*



2. *Test suite*



3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Music Playlist Batch Creator – Test Plans

USER STORIES

As a user, I want to sign up and log in securely so that I can access my playlists (ID: 79).

As a user, I need to see my playlist in one place (ID: 76).

As a user, I should be able to create an audio playlist as needed (ID: 73).

As a user, I should be able to rename, record, and change the playlist (ID: 68).

As a user, I need to have real-time metadata (ID: 65).

Test Suites

Test Suit: TS01 - User Login (ID: 86)

1. TC01 – Successful Sign Up

o Action:

- ☐ Go to the Sign-Up page.
- ☐ Enter valid name, email, and password.
- ☐ Click "Sign Up".

o Expected Results:

- ☐ Sign-Up form is displayed.
- ☐ Fields accept values without error.
- ☐ Account is created, and the user is redirected to the dashboard.

o Type: Happy Path

2. TC02 – Secure Login

o Action:

- ☐ Go to the Login page.
- ☐ Enter valid email and password.
- ☐ Click on "Login".

o Expected Results:

- ☐ Login form is displayed.
- ☐ Fields accept data without error.
- ☐ User is logged in and redirected to the dashboard.

o Type: Happy Path

3. TC03 – Sign Up with Existing Email

o Action:

- ☐ Go to the Sign-Up page.
- ☐ Enter a name and an already registered email.
- ☐ Click on "Sign Up".

o Expected Results:

- ☐ Fields accept data.
- ☐ Error message "Email already registered" is displayed.
- o **Type:** Error Path

4. TC04 – Login with Wrong Password

- o **Action:**
 - ☐ Go to the Login page.
 - ☐ Enter valid email and incorrect password.
 - ☐ Click on "Login".
- o **Expected Results:**
 - ☐ Input is accepted.
 - ☐ Error message "Invalid username or password" is shown.
- o **Type:** Error Path

Test Suit: TS02 - View Playlists (ID: 87)

1. TC05 – View Playlist Page

- o **Action:**
 - ☐ Log in successfully.
 - ☐ Navigate to "My Playlists" section.
- o **Expected Results:**
 - ☐ All created playlists are displayed clearly.
- o **Type:** Happy Path

2. TC06 – Playlist Loading Failure

- o **Action:**
 - ☐ Disconnect from the internet.
 - ☐ Navigate to "My Playlists".
- o **Expected Results:**
 - ☐ Network is offline.
 - ☐ Error message "Unable to load playlists" is shown.
- o **Type:** Error Path

Test Suit: TS03 - Real-Time Metadata (ID: 88)

1. TC07 – Real-Time Metadata Display

- o **Action:**
 - ☐ Play a song.
 - ☐ Observe the metadata panel.
- o **Expected Results:**
 - ☐ Metadata (title, artist, album, duration) is displayed and updates in real time.
- o **Type:** Happy Path

2. TC08 – Metadata Not Updating

- o **Action:**

- ☐ Play a different song.
 - ☐ Observe the metadata panel.
- **Expected Results:**
 - ☐ Metadata remains static or shows default/fallback message.
- **Type:** Error Path

Test Suit: TS04 - Playlist Editing (ID: 89)

1. TC09 – Rename Playlist Successfully

- **Action:**
 - ☐ Navigate to "My Playlists".
 - ☐ Click "Rename" next to a playlist.
 - ☐ Enter a new name and click "Save".
- **Expected Results:**
 - ☐ Playlist name updates successfully.
- **Type:** Happy Path

2. TC10 – Rename with Blank Name

- **Action:**
 - ☐ Click "Rename" on a playlist.
 - ☐ Leave the field blank.
 - ☐ Click "Save".
- **Expected Results:**
 - ☐ Error message "Playlist name cannot be empty" is shown.
- **Type:** Error Path

3. TC11 – Change Playlist Order

- **Action:**
 - ☐ Open a playlist.
 - ☐ Drag and drop songs to reorder.
 - ☐ Click "Save".
- **Expected Results:**
 - ☐ Playlist order is updated and saved.
- **Type:** Happy Path

4. TC12 – Change Playlist Order Fails

- **Action:**
 - ☐ Login and go to "My Playlists".
 - ☐ Select a playlist.
 - ☐ Go offline or simulate server error.
 - ☐ Reorder songs and click "Save Order".
- **Expected Results:**
 - ☐ Error message: "Failed to update order. Please check your connection".
- **Type:** Error Path

Test Suit: TS05 - Smart Playlist Creation (ID: 90)

1. TC13 – Generate Playlist Based on Various Categories

o **Action:**

- ☐ Login with valid credentials.
- ☐ Click on "Generate Playlist".
- ☐ Select categories.
- ☐ Click "Generate Playlist".

o **Expected Results:**

- ☐ Playlist is generated based on selected mood and categories.

o **Type:** Happy Path

2. TC14 – Fail to Generate Playlist Due to Missing Category Selection or Invalid Input

o **Action:**

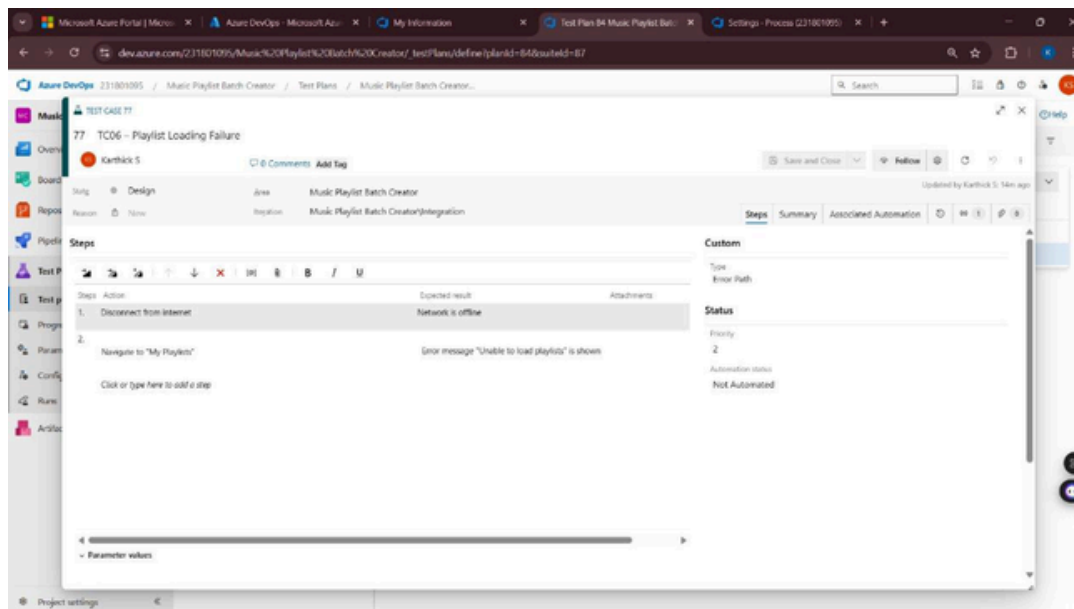
- ☐ Login with valid credentials.
- ☐ Click on "Generate Playlist".
- ☐ Select categories.
- ☐ Click "Generate Playlist".

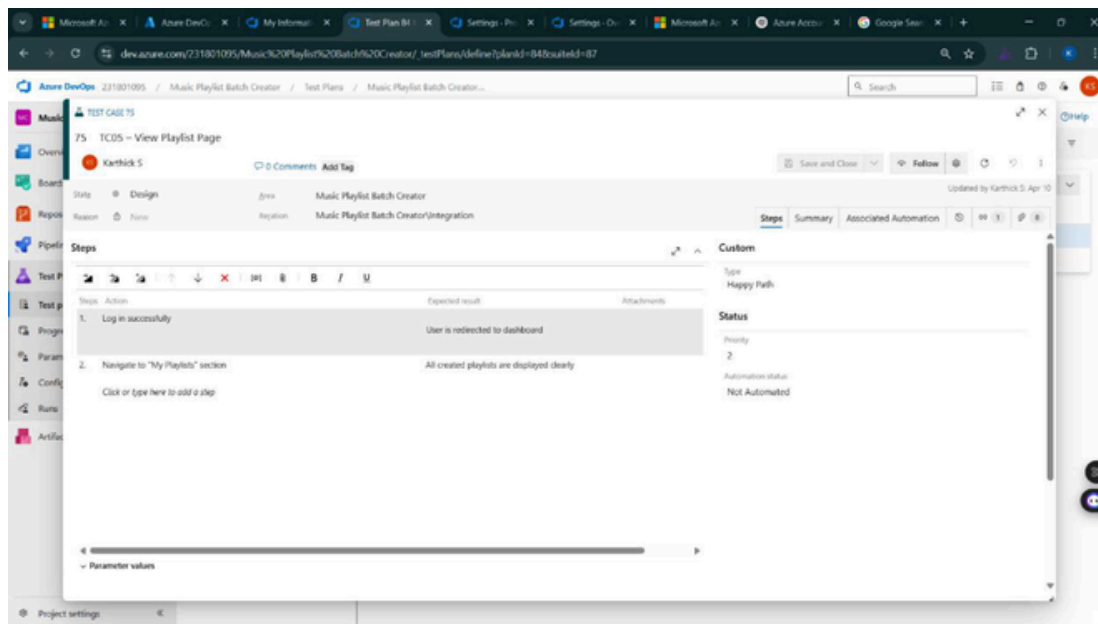
o **Expected Results:**

- ☐ Error message: "Please select at least one valid category" or "No recommendations found for the selected filters".

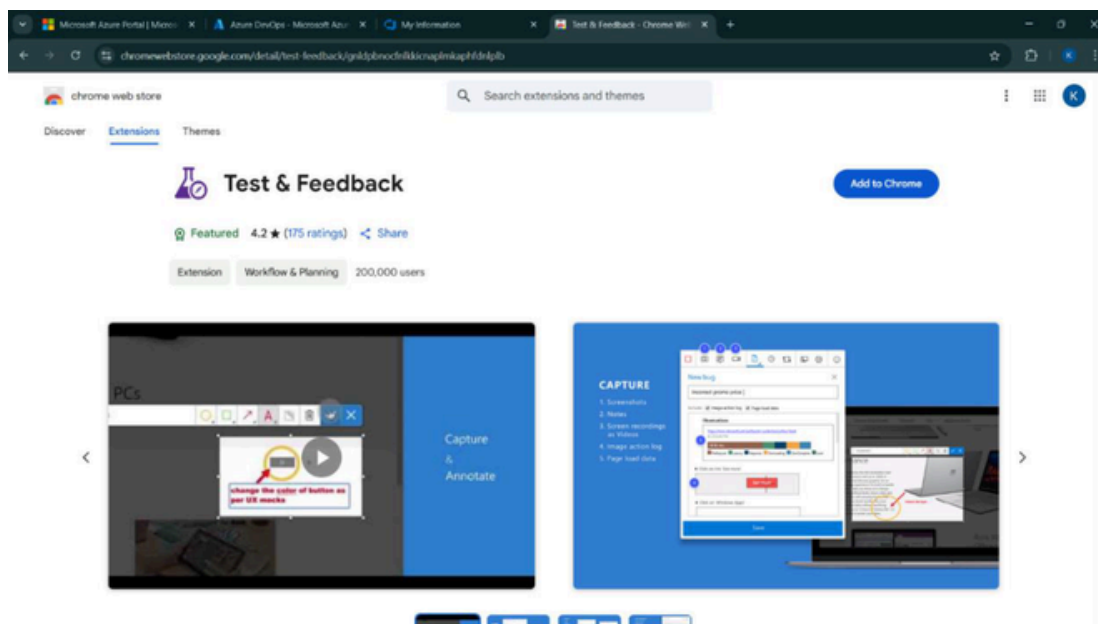
o **Type:** Error Path

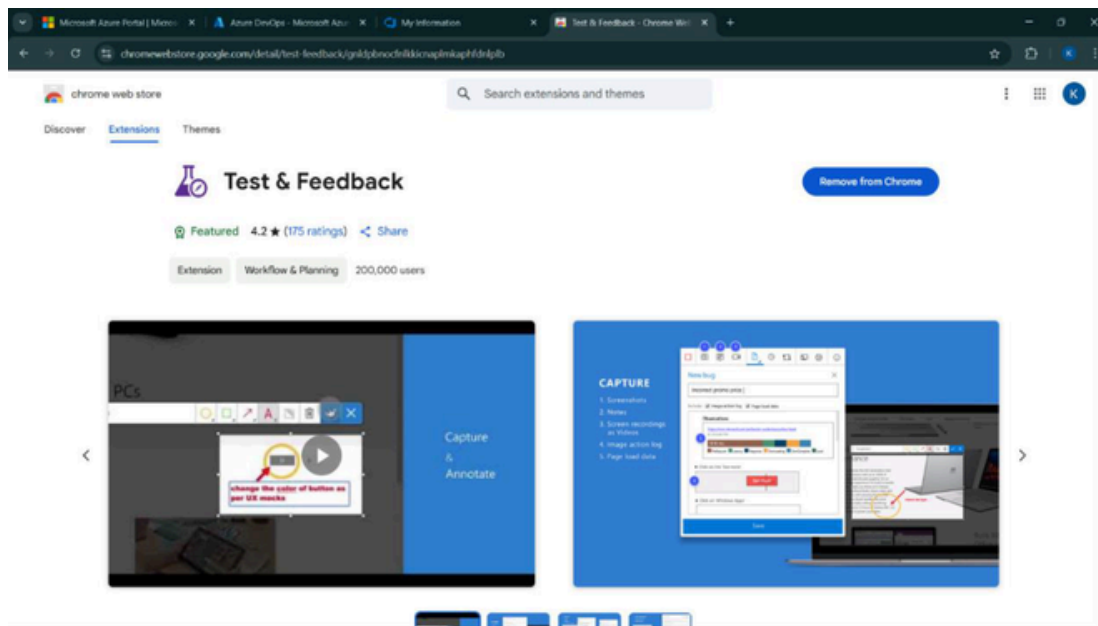
Test Cases



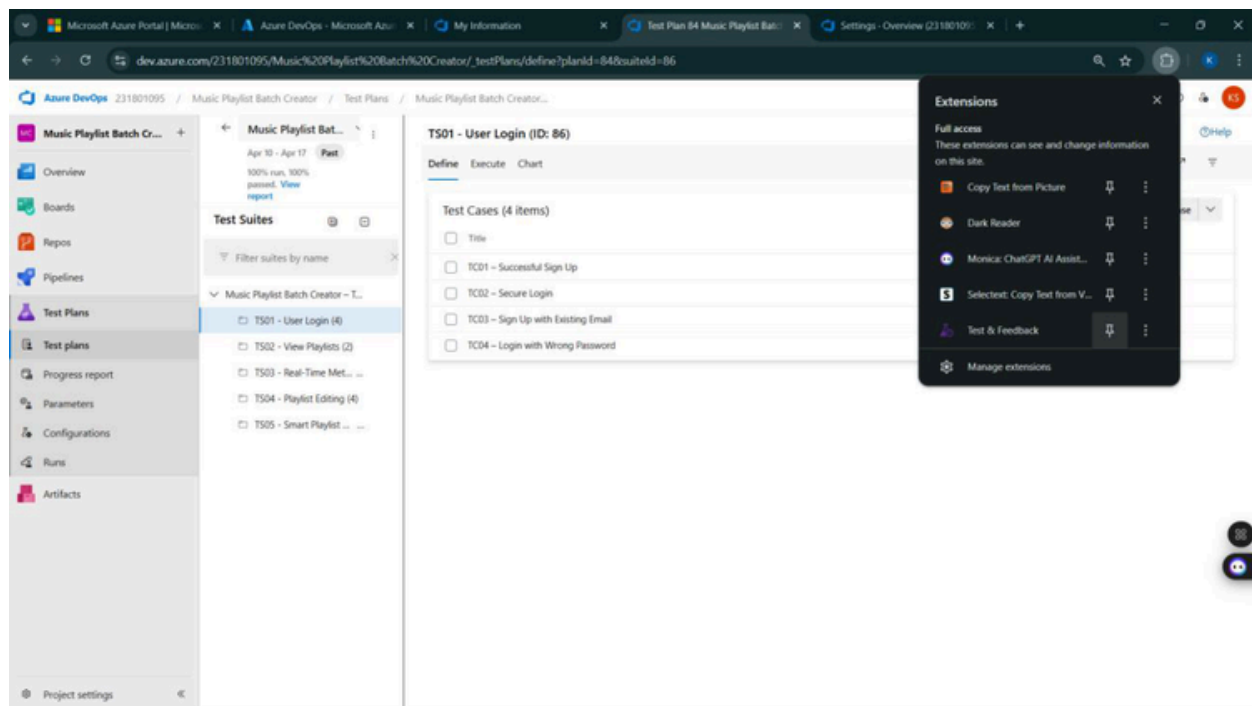


4. Installation of test





Test and feedback
Showing it as an extension



5. Running the test cases

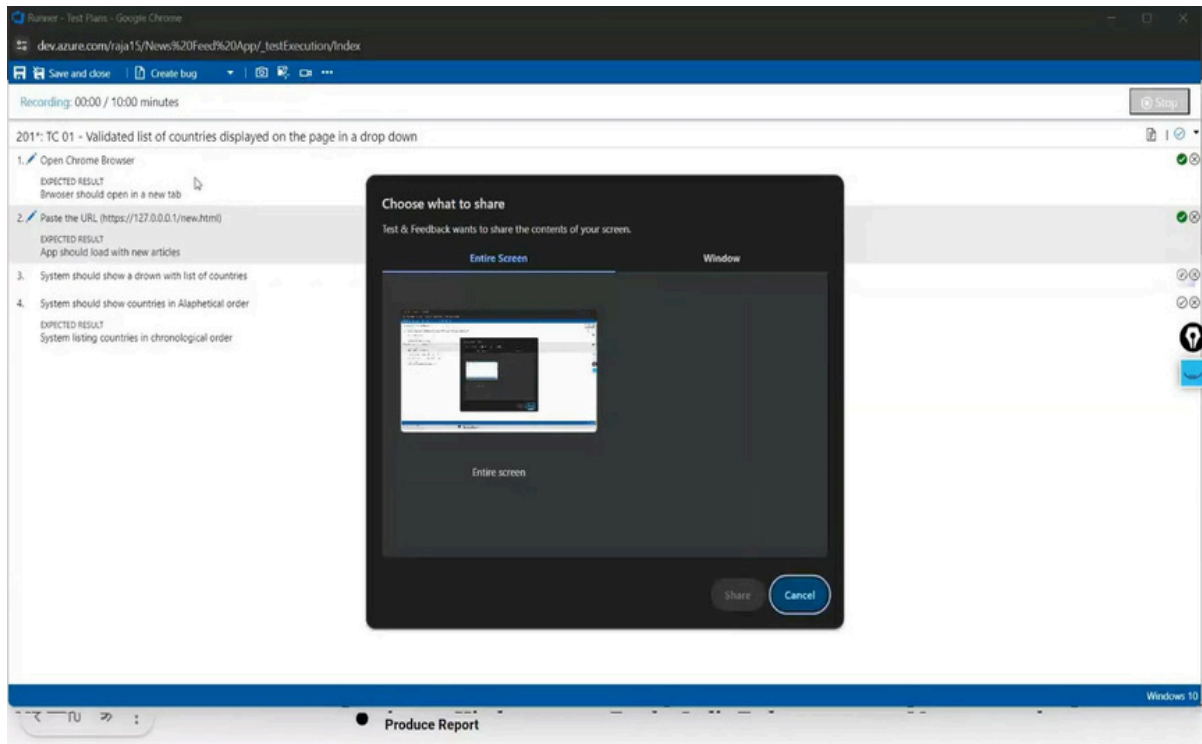
The screenshot displays the Azure DevOps Test Plans interface. The left sidebar shows the navigation menu with 'Test Plans' selected. The main area shows the 'Music Playlist Batch Creator' test plan with a 'Past' status. The 'Test Suites' section lists several suites, including 'TS02 - View Playlists (2)'. The 'Test Points' table shows two items: 'TC05 - View Playlist Page' and 'TC06 - Playlist Loading Failure'. A context menu is open for 'TC05', showing options like 'View execution history', 'Mark Outcome', 'Run', 'Reset test to active', 'Edit test case', 'Assign tester', and 'View test result'. The 'Run' option is selected, and a sub-menu shows 'Run for web application', 'Run for desktop application', and 'Run with options'.

Title	Outcome	Order	Test Case Id	Configuration	Tester
TC05 - View Playlist Page	Passed	1	75	Windows 10	Malu karthick B...
TC06 - Playlist Loading Failure	Passed	2	77	Windows 10	Malu karthick B...

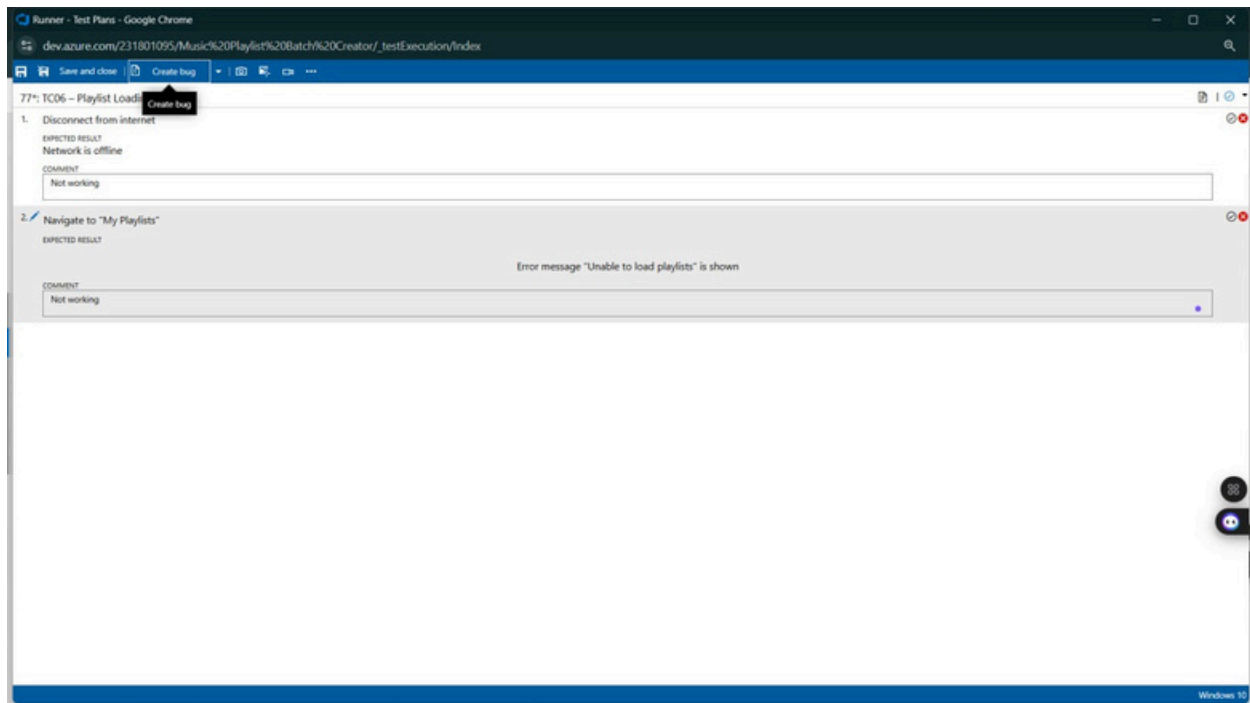
The screenshot displays the Azure DevOps Test Results page for the 'Music Playlist Batch Creator' test plan. The page shows the execution of two test cases: '75: TC05 - View Playlist Page' and '77: TC06 - Playlist Loading Failure'. The results are as follows:

Test Case	Step	Expected Result	Actual Result
75: TC05 - View Playlist Page	1. Log in successfully	User is redirected to dashboard	Passed
	2. Navigate to "My Playlists" section	All created playlists are displayed clearly	Passed
77: TC06 - Playlist Loading Failure	1. Load playlist	Playlist is loaded successfully	Failed
	2. Verify playlist loading	Playlist is not loaded	Failed

6. Recording the test case



7. Creating the bug



Runner - Test Plans - Google Chrome

dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_testExecution/index

77% TC06 - Playlist Loading Failure

1. Disconnect from Internet

NEW BUG

TB01 - Playlist loading spinner keeps spinning indefinitely on poor network

Unassigned 0 comments Add tag TB01 - Playlist loading spinner keeps spinning indefinitely on poor network Save & Close

2. Navigate

Expected Result

Music Playlist Batch Creator

Music Playlist Batch Creator

Details

Repro Steps

18-04-2025 09:23 Bug filed on "TC06 - Playlist Loading Failure"

Step no. Result Title

1. Failed Disconnect from internet

Expected Result

Network is offline

Comments: Page Not loading

2. Failed Navigate to "My Playlists"

Expected Result

Error message "Unable to load playlists" is shown

Test Configuration: Windows 10

Planning

Resolved Reason

Story Points

Priority

2

Severity

3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

+ Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

+ Add link

Add an existing work item as a parent

Tested By

77 TC06 - Playlist Loading Failure

Updated 10-04-2025, @ Design

System Info

Found in Build

Microsoft Azure Portal

Azure DevOps - Music Playlist Batch Creator

My Information

Test Plan 84 Music Playlist Batch Creator

Runs - Test Plans

Settings - Overview

Bug Report Playlist

dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_testManagement/runs?_a=resultSummary&runId=48&resultId=100000

Azure DevOps 231801095 / Music Playlist Batch Creator / Test Plans / Runs

92 TB01 - Playlist loading spinner keeps spinning indefinitely on poor network

Unassigned 0 comments Add tag TB01 - Playlist loading spinner keeps spinning indefinitely on poor network Save & Close Follow

Updated by Karthick S. 8m ago

Details

System Info

Found in Build

Integrated in Build

System Info

Browser - Name	Google Chrome 135
Browser - Language	en-IN
Browser - Height	864
Browser - Width	1536
Browser - User agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36
Operating system - Name	Windows NT 10.0; Win64; x64
Operating system - Architecture	x64_64
Operating system - Processor	11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz
Operating system - Number of processors	4
Memory - Available	814784512
Memory - Capacity	8214240128
Display - Pixels per inch (X axis)	120
Display - Pixels per inch (Y axis)	120
Display - Device pixel ratio	1.25

Discussion

Add a comment. Use # to link a work item, / to link a pull request, or @ to mention a person.

Project settings

Name

SystemInformation-2025-04-18T01-23-58.168Z.json

Size

1K

8. Test case results

The screenshot shows the Azure DevOps interface for a project named 'Music Playlist Batch Creator'. The 'Test Plans' section is active, displaying a list of test plans. The 'TS02 - View Playlists (ID: 87)' test plan is selected, showing its 'Test Points' and 'Test Case Results'.

Test Points (2 items):

- TC05 - View Playlist Page
- TC06 - Playlist Loading Failure

Test Case Results:

Outcome	Timestamp	Configuration	Run by	Tester	Test Plan
Passed	4m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	12m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Not Applicable	12m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	14m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	Tuesday	Windows 10	Karthikayan Senthil	Mallu karthick Balaji ...	Music
Passed	Saturday	Windows 10	Mallu karthick Balaji ...	Mallu karthick Balaji ...	Music
Failed	Saturday	Windows 10	Mallu karthick Balaji ...	Mallu karthick Balaji ...	Music
Passed	Apr 11	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	Apr 11	Windows 10	Karthick S	Mallu karthick Balaji ...	Music

9. Test report summary

The screenshot shows the Azure DevOps interface for a project named 'News Feed App'. The 'Work Items' section is active, displaying a list of work items. The 'BUG 203 - BG 01 - Countries Drop down Not Available on the page' work item is selected, showing its details and test results.

BUG 203 - BG 01 - Countries Drop down Not Available on the page

State: New
Reason: New
Repro Step: Active
5/19/2021 AM

Test Results:

Step no.	Result	Title
1.	Passed	Open Chrome Browser
2.	Passed	Paste the URL (https://127.0.0.1/new.html)
3.	Failed	Section should show a dropdown with list of countries

- Assigning bug to the developer and changing state

The screenshot shows the Azure DevOps Test Plan interface for a project named 'Music Playlist Batch Creator'. The selected test plan is 'TC06 - Playlist Loading Failure'. The test run status is 'Failed'.

Repro Steps:

- Step no. 1:** Result: Failed. Title: Disconnect from internet. Expected Result: Network is offline. Comments: Page Not loading.
- Step no. 2:** Result: Failed. Title: Navigate to "My Playlists". Expected Result: Error message "Unable to load playlists" is shown.

Test Configuration: Windows 10

Planning: Resolved Reason, Story Points, Priority 2, Severity 3 - Medium, Activity.

Deployment: To track releases associated with this work item, go to Releases and turn on deployment status reporting for Branches in your pipeline's Options menu. Learn more about deployment status reporting.

Development: Add link. Link an Azure Repos commit, pull request, or branch to see the status of your development. You can also create a branch to get started.

Related Work: Add link. Add an existing work item as a parent.

Tested By: 77 TC06 - Playlist Loading Failure. Updated 10-04-2025. Design.

System Info:

10. Progress report

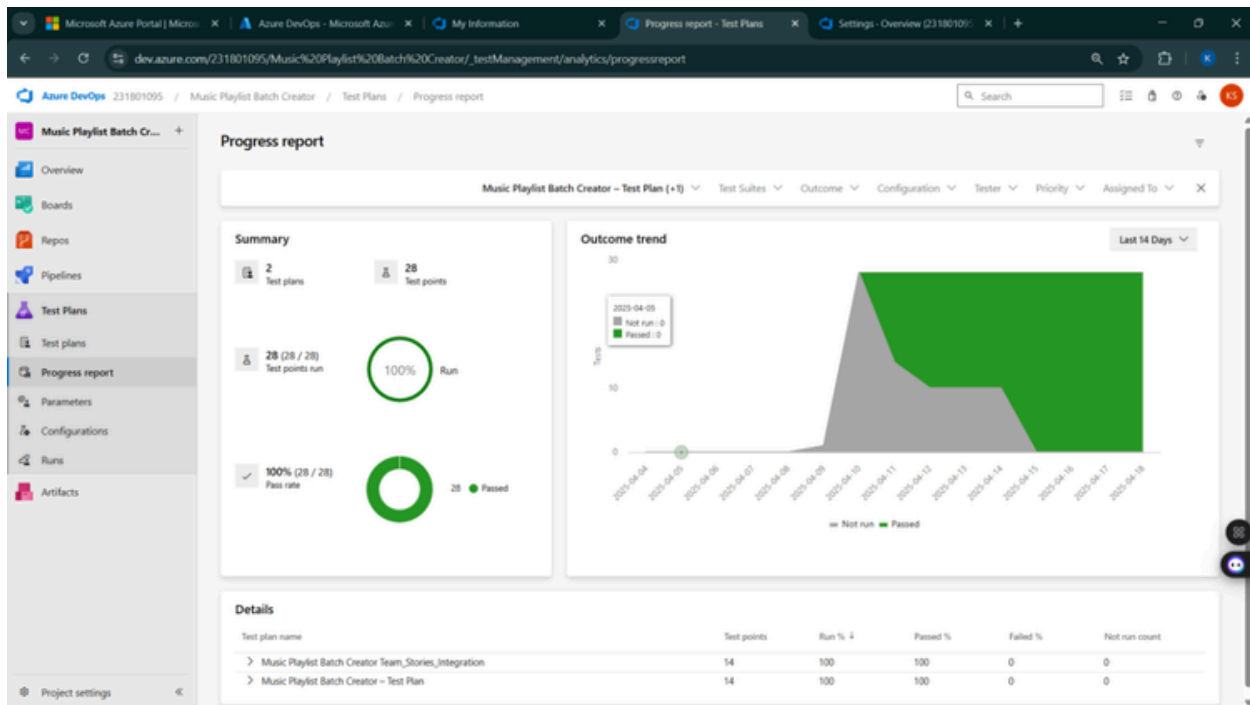
The screenshot shows the Azure DevOps Test Plan interface for a project named 'Music Playlist Batch Creator'. The selected test plan is 'Music Playlist Batch Creator - Test Plan'. The progress report shows the following summary:

- 1 Test plan
- 14 Test points
- 14 (14 / 14) Test points run
- 100% Run
- 100% (14 / 14) Pass rate
- 14 Passed

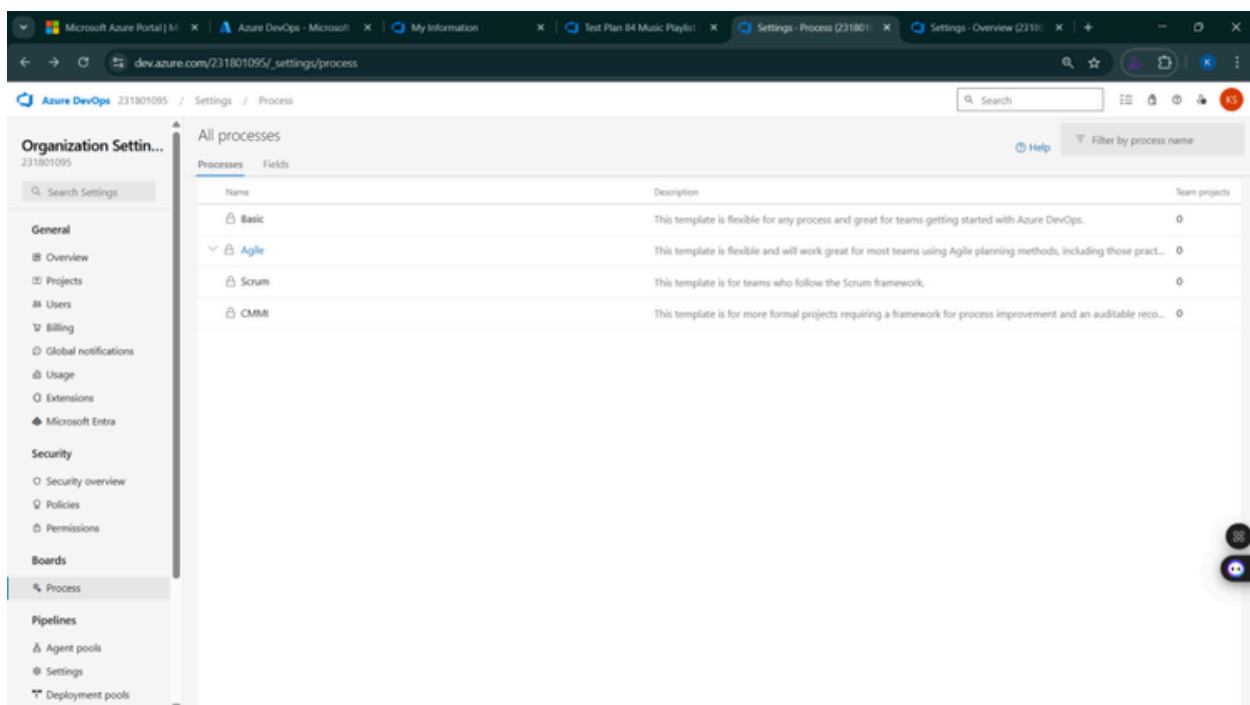
Outcome trend: Last 14 Days. The chart shows a sharp increase in the number of test points run starting around 2025-04-08, reaching a peak of 14 test points by 2025-04-10, and remaining at that level through 2025-04-15.

Details:

Test plan name	Test points	Run %	Passed %	Failed %	Not run count
Music Playlist Batch Creator - Test Plan	14	100	100	0	0
TS01 - User Login	4	100	100	0	0
TS02 - View Playlists	2	100	100	0	0
TS03 - Real Time Metadata	2	100	100	0	0
TS04 - Playlist Editing	4	100	100	0	0
TS05 - Smart Playlist Creation	2	100	100	0	0



11.Changing the test template



Microsoft Azure Portal | M | Azure DevOps - Microsoft | My Information | Test Plan B4 Music Player | Settings - Process (231801095) | Settings - Overview (231801095) | dev.azure.com/231801095/_settings/process

Azure DevOps 231801095 / Settings / Process

Organization Settings 231801095

Search Settings

General

- Overview
- Projects
- Users
- Billing
- Global notifications
- Usage
- Extensions
- Microsoft Entra

Security

- Security overview
- Policies
- Permissions

Boards

- Process

Pipelines

- Agent pools
- Settings
- Deployment pools

All processes

Processes Fields

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record.	0

Microsoft Azure Portal | M | Azure DevOps - Microsoft | My Information | Test Plan B4 Music Player | Settings - Process (231801095) | Settings - Overview (231801095) | dev.azure.com/231801095/_settings/process

Azure DevOps 231801095 / Settings / Process

Organization Settings 231801095

Search Settings

General

- Overview
- Projects
- Users
- Billing
- Global notifications
- Usage
- Extensions
- Microsoft Entra

Security

- Security overview
- Policies
- Permissions

Boards

- Process

Pipelines

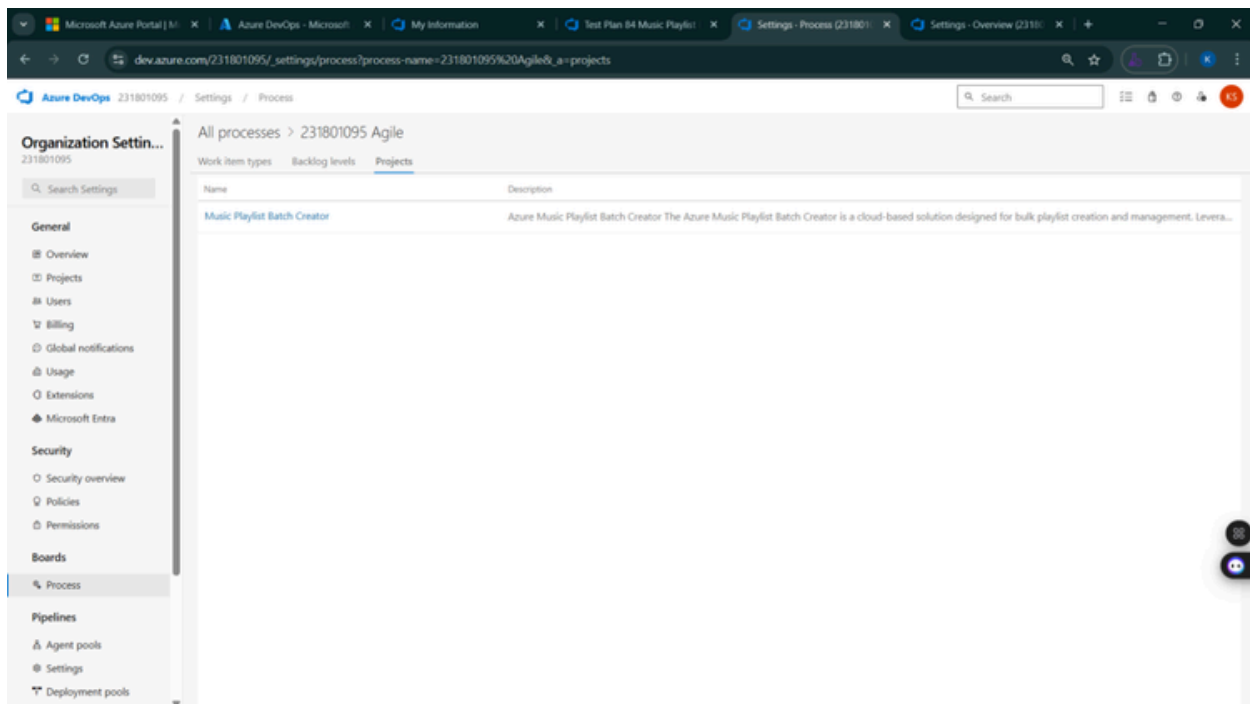
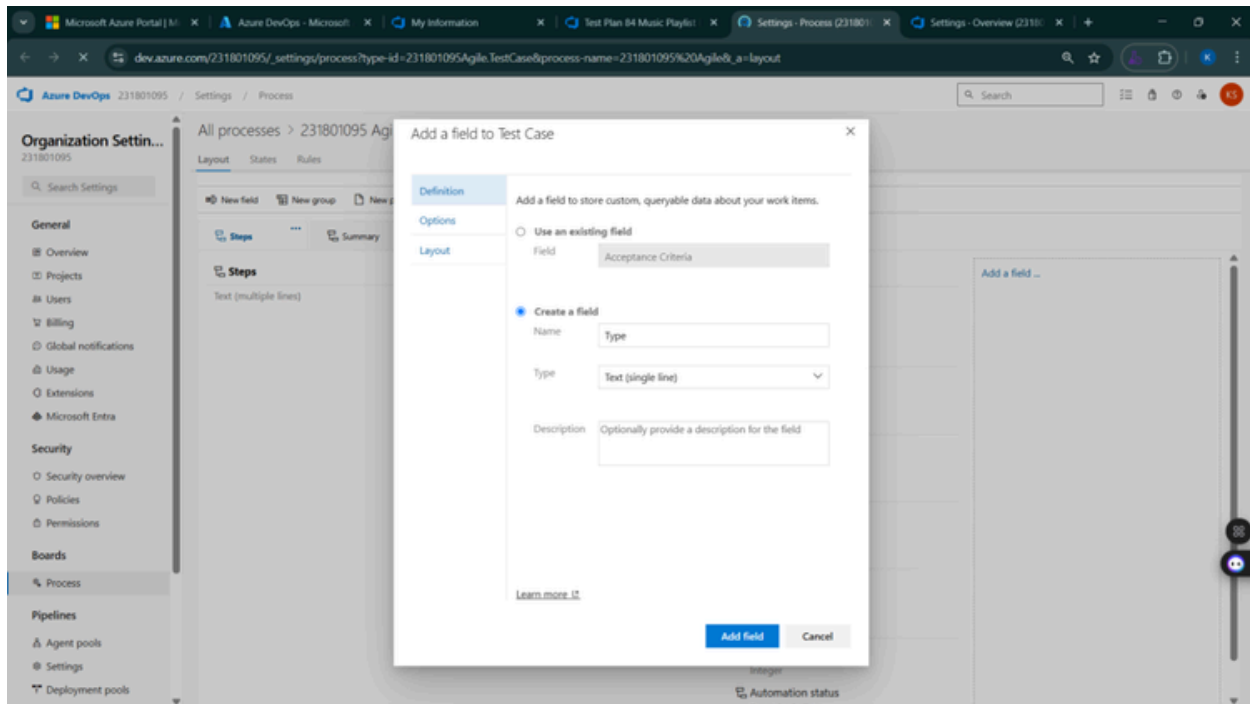
- Agent pools
- Settings
- Deployment pools

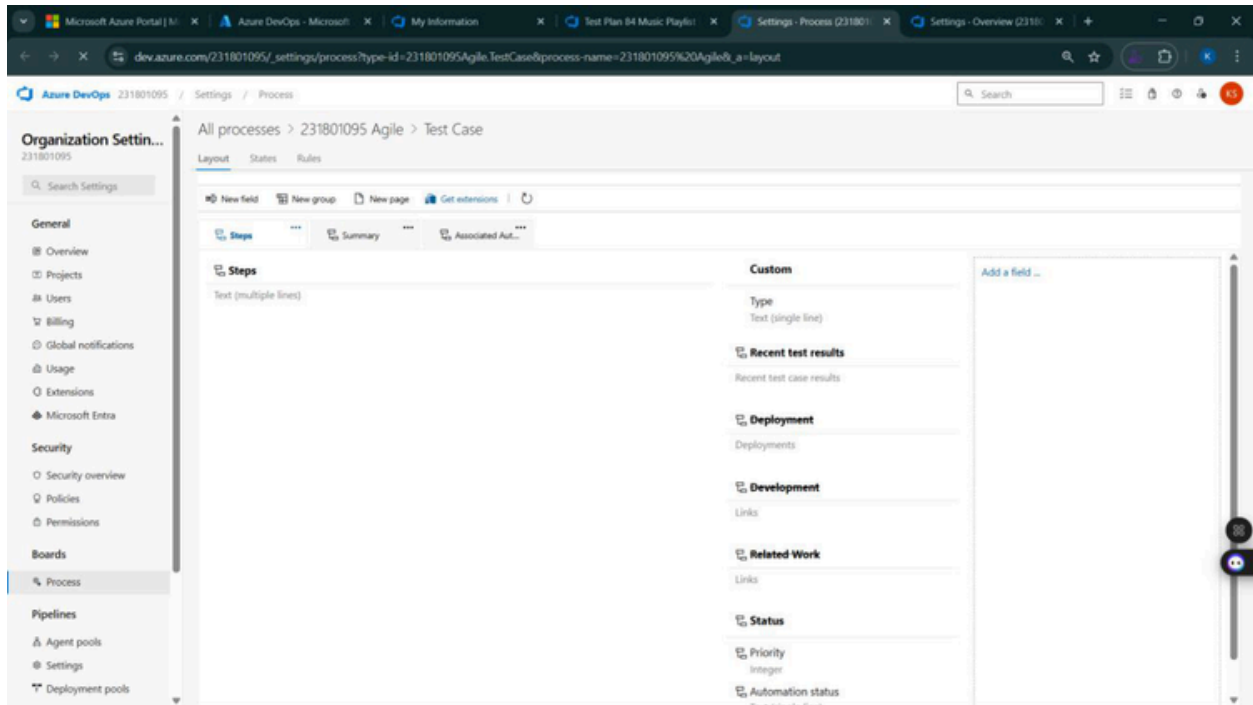
All processes

Processes Fields

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	0
231801095 Agile (default)		1
Agile Plus		0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record.	0

12. View the new test case template





Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

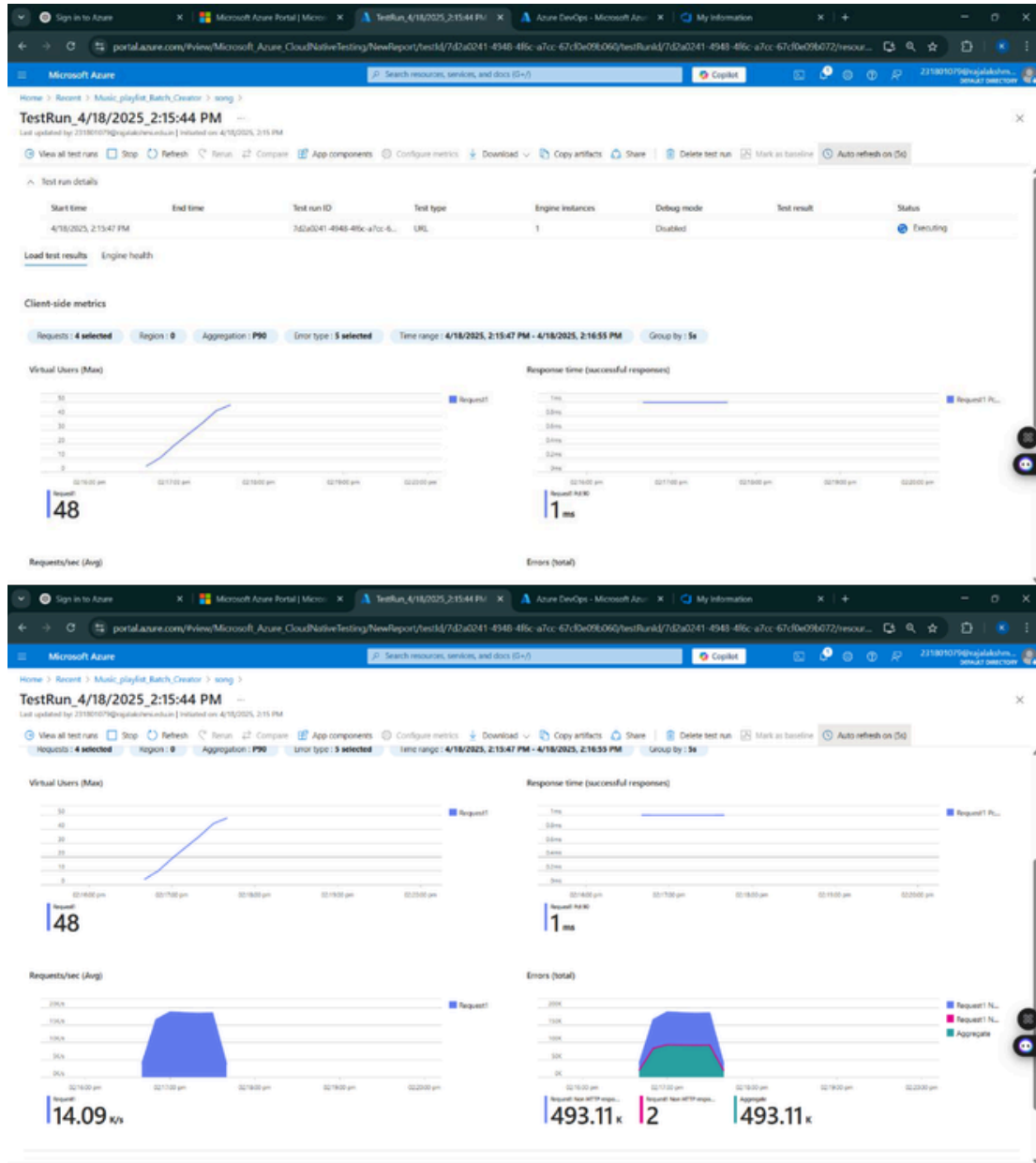
1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource
 - o Go to **Create a resource** → Search for “Azure Load Testing”.
 - o Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
 - o **Subscription:** Choose your Azure subscription.
 - o **Resource Group:** Create new or select an existing one.
 - o **Name:** Provide a unique name (no special characters).
 - o **Location:** Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

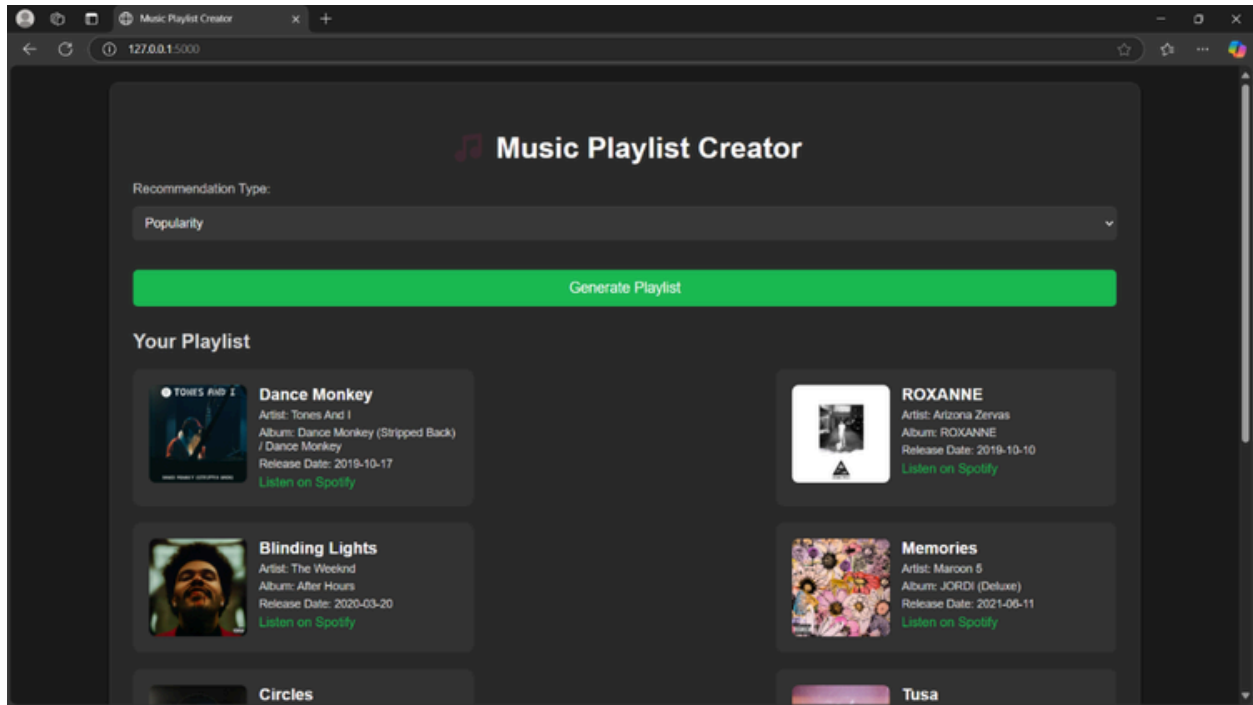
Steps to Create and Run a Load Test:

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab
 - o **Test Name:** Provide a unique name.
 - o **Description:** (Optional) Add test purpose.
 - o **Run After Creation:** Keep checked.
3. Load Settings
 - o **Test URL:** Enter the target endpoint (e.g., <https://yourapi.com/products>).
4. Click Review + Create → Create to start the test.

Load Testing





Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted Flask-based music recommendation project with Azure DevOps. The pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
 - o In Azure DevOps, create a new project.
 - o Create a pipeline and select GitHub as the source.
 - o Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
 - o In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - o Add the following basic pipeline configuration for Python and Flask:

yml Code

trigger:

-main # Trigger pipeline when changes are pushed to the main branch

pool:

vmImage: ubuntu-latest # Use a hosted Ubuntu agent

steps:

#Step 1: Checkout the code from GitHub

-checkout: self

#Step 2: Set up Python environment

-task: UsePythonVersion@0

inputs:

versionSpec: '3.x' # Use the latest Python 3.x version

displayName: "Set up Python"

#Step 3: Install dependencies from the correct path

-script: |

python -m pip install --upgrade pip

pip install -r project/requirements.txt # Adjusted path to requirements.txt

displayName: "Install dependencies"

#Step 4: Run a simple Python script to check the environment

-script: |

python -c "print(' Hello from Music Playlist Batch Creator!')"

displayName: "Run a Python script"

3. Pipeline Tasks Include:

- o Setting up the Python environment using the UsePythonVersion task.
- o Installing project dependencies from project/requirements.txt. Make sure the path to requirements.txt is correct (it is located under the project folder).
- o Running a simple Python script to verify that Python is set up correctly and the pipeline works.

4. Run and Monitor Pipeline:

- o Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
- o Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline

The screenshot displays the Azure DevOps web interface. The left sidebar contains navigation links for Overview, Boards, Repos, Pipelines, Environments, Releases, Library, Task groups, Deployment groups, Test Plans, and Artifacts. The main area shows the details of a pipeline run labeled '#20250424.3 • Pipeline 2' for the 'Music Playlist Batch Creator' project. A message indicates the run is retained as one of 3 recent runs by main (branch). The 'Summary' tab is active, showing it was manually run by 'Karthick S' at 'Just now' (24s). It lists repository and version details, related work items, and tests and coverage. Below this, a 'Jobs' table shows a single job named 'Job' with a status of 'Success' and a duration of 6s. The 'View 52 changes' link is also visible.

Name	Status	Duration
Job	Success	6s

Result:

Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

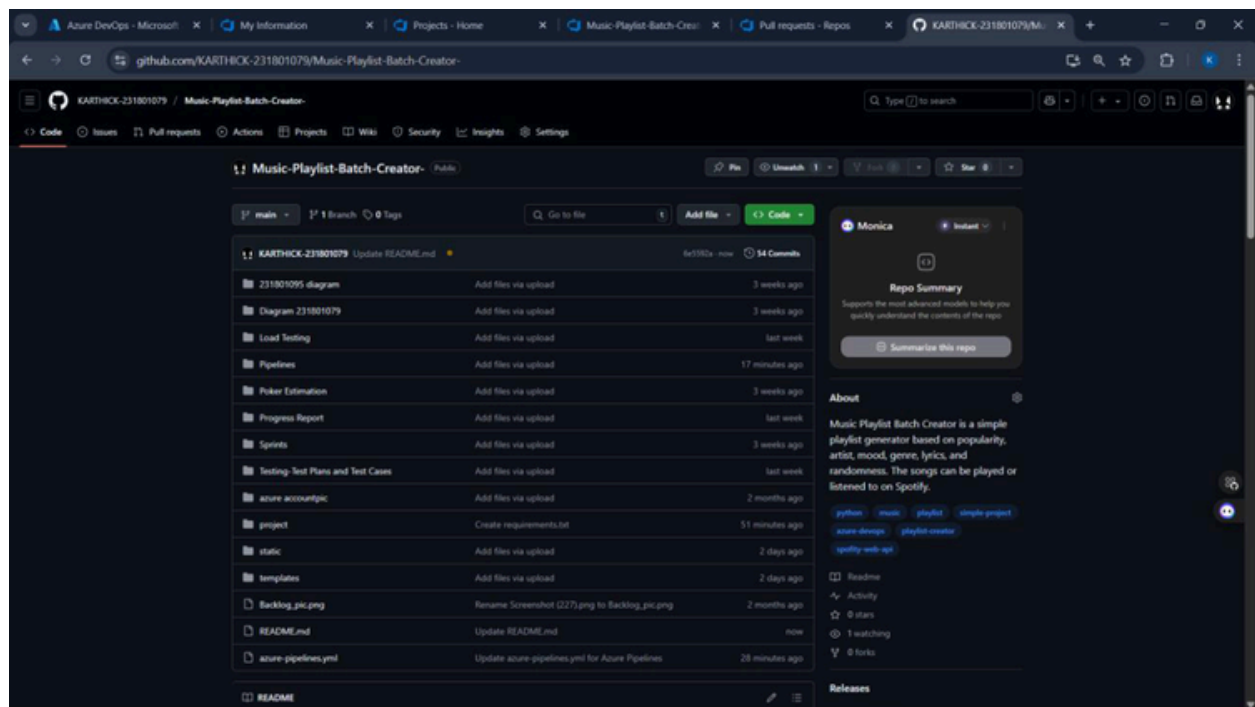
EXP NO: 10

GITHUB:PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Music Playlist Batch Creator project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.